

Johnathan Gilday

Distinguished Engineer building the agents that fix vulnerabilities before they reach production. 15+ years experience. Presently, at the intersection of applied AI and application security: designing agent harnesses, sub-agent architectures, and human-in-the-loop workflows that let AI autonomously validate and remediate software vulnerabilities. Founding-team contributor at venture-backed AI-native startups, former cleared engineer supporting mission-critical systems for the U.S. Intelligence Community, and an active voice driving organizational AI transformation across engineering.

EXPERIENCE

Pixee — *Distinguished Engineer*

DECEMBER 2022 - PRESENT

- **Founding-team technical strategist** for Pixee's AI-native application security platform, serving Fortune 500 and Global 3000 enterprises across financial services, fintech, and technology—designing the agent harnesses, sub-agent architectures, and human-in-the-loop workflows that let AI autonomously validate and remediate vulnerabilities.
- **Built the agentic security stack hands-on:** MCP tool integrations, context engineering, and multi-agent orchestration that turn vulnerability backlogs into reviewed, merged pull requests.
- **Led the architectural pivot** from AWS-serverless multi-tenant SaaS to a Helm-based on-premises deployment via Replicated, unlocking enterprise adoption while sustaining SaaS delivery from a unified codebase.
- **Drive organizational AI transformation**, rolling out Claude Code agents and agentic engineering workflows across the org, and publishing thought leadership on agentic coding and the AI-native SDLC at johnthangilday.com.

Contrast Security — *Distinguished Engineer*

MARCH 2017 - DECEMBER 2022

- **Owned technical direction** for Contrast's flagship Java agent, the company's primary revenue driver, through hypergrowth from Series B to Series E—delivering real-time, in-production security analysis via

Red Bank, NJ
me@johnthangilday.com
<https://johnthangilday.com>
[Blog](#)
[GitHub](#)
[Stack Overflow](#)
[Baeldung Articles](#)

EDUCATION

The Johns Hopkins University —
M.S. Computer Science

Rutgers University — B.S.
Computer Science

CORE COMPETENCIES

Agentic AI & LLM Engineering

Agent harness design, sub-agent architectures, multi-agent orchestration, context engineering, prompt engineering, MCP tool integrations

Applied AI Security

Autonomous vulnerability remediation, human-in-the-loop agent workflows, AI-native AppSec platforms

Backend Architecture & Performance

JVM ecosystem, distributed systems, CI/CD, containerization, scalable system design

Customer-Facing Technical Leadership

Stakeholder communication,

JVM bytecode transformation, including engineering the JNDI injection detection rule in Contrast Protect that generalized real-time defense against Log4Shell to the broader vulnerability class.

- **Optimized performance-critical analysis** running inside customer production systems, reducing lock contention and introducing caching and lazy evaluation.
- **Redesigned the integration-testing system**, cutting feedback cycles from weeks to daily for 24+ developers and scaling coverage to 350+ containerized test applications.
- **Owned customer-facing technical work**: sales engineering for enterprise prospects and resolution of escalated support issues.
- **Set long-term technical vision** as the organization scaled past 100 engineers, mentoring widely (including book clubs) as the company's top individual-contributor rank.

Next Century Corporation — Senior Software Engineer

NOVEMBER 2012 - MARCH 2017

- **Built greenfield systems** for U.S. Department of Defense programs, including cross-domain solutions and a customer data-filtering system delivering user log analysis, with cloud automation via Apache NiFi and Python.
- **Maintained a mission-critical data API** serving an enterprise microservice mesh, raising reliability through expanded test coverage and automated deployment pipelines with production monitoring.

U.S. Army CERDEC — Computer Scientist

JULY 2007 - November 2012

Developed prototype mission control applications for web, mobile, and multi-touch platforms, co-authoring [award-winning research on emerging interface technologies](#).

sales engineering, customer escalation resolution, technical product strategy

Cross-Functional Leadership

Team mentoring, technical vision, organizational AI transformation, process improvement

AWARDS

2017 Above and Beyond the Call of Duty Award at Contrast Security

Multiple team and early-career awards from U.S. Army CERDEC, PEO C3T, AFCEA, and CompTIA recognizing innovation and technical excellence

US DOD Security Clearance

TS//SCI Poly: 2013-03-27
INDOC Date: 2013-04-23
Debrief: 2017-03-03

DJ2-2016-03389